Is this reading the one the wall was expected to show?

A grey-box NeuralProphet decomposition, rolling expectation and three-scale monitor of the station 02 inclination, 2018 to 2026

Study report · `studies/05_greybox_monitoring/`

Contents

1 Introduction

A reading arrives from the wall every twenty minutes: the inclination of the instrument at station 02, the air temperature and relative humidity beside it, and, since 21 February 2025, the temperature of the stone and the solar radiation falling on it. The question a person watching that stream must be able to answer is operational rather than diagnostic. Given the environment measured at this instant, is this inclination the one the wall was expected to show, and if it is not, which kind of departure is it? Study 04 posed that question on the window after the long outage of 2022 to 2023 and answered part of it. It also recorded its own weak points: an annual term that was not identifiable on 3.2 interrupted cycles, and a trend that changed sign across the thirds of its window; a conformal interval calibrated once, eleven months before the record it was then asked to cover, that reached 67.7 % coverage against a nominal 90 %; a residual whose lag-one autocorrelation of 0.995 pushed the control limit to 14.75 standard deviations and left the detector blind to every slow mechanism; and a phase-change injection scaled by the leftover daily seasonal term rather than by the wall's own daily response, which overstated that blind spot by roughly five times. This study keeps what Study 04 established and changes what it got wrong.

Four studies precede it, and their conclusions are inherited rather than re-litigated. Study 01 read the raw archive, decided which recorded values are measurements, and exported the inclination product used here and no other: compensated for temperature with the manufacturer's linear correction, anchored once across the whole record, cleaned of impulsive noise, and accompanied by a flag that marks every value the cleaning replaced. It also established that the two instruments installed at station 02 are continuous across the changeover of 21 February 2025, and that the excursions of summer 2026 are an instrument or power artefact rather than a structural or environmental event. Study 02 characterised the two external sources, the town's ground station and the ERA5 reanalysis, against the on-structure channels: the station is the closer proxy, ERA5 the more transferable one, and both report global horizontal radiation, the quantity the wall's own pyranometer measures. Study 03 measured the couplings: air temperature drives the inclination instantaneously, with a diurnal-band gain near -2.8 mdeg/°C on the on-structure channel; radiation acts with a delay of about one hour; humidity follows the temperature cycle inversely rather than acting as a mechanism of its own; and the wall temperature is an internal state that leads the deformation and must not be used predictively at its measured lead. Study 04 settled the modelling choices this study keeps: no autoregression in the decomposition, because it absorbs the diurnal structure the regressors should explain; changepoints placed on covered time; the level decomposed and monitored rather than scored as a forecast; contemporaneous regressors in the expectation treated as legitimate inputs rather than as leakage; and no value of the target ever imputed, with the library's own gap filling switched off.

Two boundaries hold throughout. The thermal compensation is taken as given: it is the manufacturer's procedure, it has been validated against the user's numerical model of the wall, and every gain this study learns is therefore a post-compensation wall response and is labelled as such. No fit is made on the raw channel. Forecasting is out of scope, because Study 04 answered it: the skill of a forecast comes from the response's own memory, and the environment adds nothing distinguishable from zero beyond six hours.

Within those boundaries the study asks five questions. What is the record made of, over eight years: trend, annual cycle, daily cycle and the response to each measured driver, with the share of variance each carries and the gain each learns, on three regressor sets? Does the wall answer its drivers with a delay or an inertia that the twenty-minute grid can resolve, when the response is learned rather than imposed? Is this reading the expected one, given a rolling expectation and an interval whose coverage is measured as a function of how stale the last refit is? Which departure does each of three control charts catch, at a false-alarm budget stated in advance, when the injected departures are sized by the wall's measured daily response? And what happened across each outage, when the level expected at resumption from the proxies that kept recording is set against the level observed?

2 The record and the three regressor sets

The study window is the whole station 02 record, from 26 July 2018 to the end of the archive in August 2026, on the instrument's native grid of one reading every twenty minutes: 212,251 slots. Unlike Study 04, which worked on the stretch after the long outage of 2022 to 2023, this study uses every usable annual cycle, because an annual term cannot be identified on three broken years, and a trend estimated on such a window disagreed with itself in sign (Section 3, D1).

The response is joined by three regressor sets, each mapping the same three roles, air temperature, relative humidity and solar radiation, to a different source (D2). The on-structure set reads air temperature and humidity from the wall's own package, joined across the two instrument eras; the station set reads the town's ground station; the ERA5 set reads the reanalysis. The on-structure set borrows the station's radiation, because the wall's pyranometer did not exist before February 2025 and Study 01 condemned 161 of its days after that; every table that reports the on-structure set carries that borrowing. The two external sources are hourly or half-hourly and are brought onto the twenty-minute grid by linear interpolation between consecutive native observations, never across a longer gap, and the ERA5 radiation, an accumulation over the preceding hour, is centred on the half-hour before it is interpolated (D13). Regressor dropouts of at most two hours are then filled by linear interpolation and flagged; the target is never filled.

Table 1 gives what each set actually holds over the window. The inclination returns an accepted value in 67.7 % of the slots. The on-structure air temperature and humidity reach 71.1 %, with about two per cent of their accepted slots filled across short dropouts; the station's channels reach 91.7 to 94.7 %, with under one per cent filled; ERA5 covers 99.2 % of the window with nothing to fill. The radiation of the on-structure and station sets is the same column and has the same coverage.

Table 1: Accepted and filled slots and coverage per regressor set and role over the study window, on the twenty-minute grid, with the target as the last row. The on-structure set's radiation is the station's column, borrowed. Read from `GM_01_window_coverage.csv`.

Set	Role	Accepted slots	Filled slots	Coverage
`str`	`tair`	150,844	3,049	71.1%
`str`	`rh`	150,844	3,045	71.1%
`str`	`sr`	201,037	1,090	94.7%
`gs`	`tair`	200,862	1,145	94.6%
`gs`	`rh`	194,675	1,140	91.7%
`gs`	`sr`	201,037	1,090	94.7%
`era5`	`tair`	210,473	0	99.2%
`era5`	`rh`	210,473	0	99.2%
`era5`	`sr`	210,455	0	99.2%
`target`	`inc_comp_cleaned`	143,795	0	67.7%

Table 2 shows how the target's missing time is distributed. Of the 3,302 gaps in the accepted inclination, 2,616 last an hour or less and 595 between one and six hours; together they hold 2,679 of the 22,819 missing hours. Nine gaps longer than seven days hold 19,234 hours, 84 % of the missing time. These are gaps in the accepted values rather than in the raw archive, and they are therefore longer than the outages the data-quality report lists: two interruptions of three weeks and one of eight in the autumn of 2018 and the first half of 2019, when the legacy instrument's readings were condemned; 57 days in the summer of 2020, around an archive outage of eleven; a single gap of 437.6 days from 9 April 2022 to 21 June 2023, in which the two outages of 2022 and the condemned weeks before and between them merge; and the outages of 2024, 2025 and

2026, of 41.5, 42.7, 17.8 and 104 days. Nothing is written into any of them. They are what the rolling expectation must be refitted around, and the ones that fall in the monitored period are the hypotheses that Section 11 tests.

Table 2: The target's missing time by gap duration: how many gaps of each class the accepted inclination has, and how many hours they hold. Read from `GM_02_gap_inventory.csv`.

Gap class	Gaps	Hours
`<=1h`	2,616	1,180
`1-6h`	595	1,499
`6-24h`	81	881
`1-7d`	1	25
`>7d`	9	19,234

Before anything was joined, each radiation source's clock was checked with Study 02's two-sided test, on the hourly grid the test was designed for (Table 3): the offset of the daily maximum from computed solar noon, and the lag of the cross-correlation against ERA5. The station and ERA5 pass both tests, with solar offsets of 0.11 and 0.44 hours and no cross-correlation lag. ERA5's larger offset is the signature of its hourly accumulation, which is stamped at the end of the hour it describes and which D13 centres before interpolation. The on-structure pyranometer fails the agreement test over its 215 certified days, its solar offset of -0.22 hours sitting against a cross-correlation lag of one hour. No regressor set consumes that channel in the main line; it enters only the current-era ladder of D4, where its clock is part of what the ladder judges.

Table 3: Study 02's clock check on each radiation source, on the hourly grid: the days with a usable solar maximum, the offset of that maximum from computed solar noon with its interquartile range, the cross-correlation lag against ERA5 with its correlation, and whether the two tests agree. Read from `GM_03_clock_check.csv`.

Source	Channel	Days	Solar offset [h]	IQR [h]	Lag [h]	r	Agree
`str`	`sr_str`	215	-0.22	0.72	1.0	0.932	no
`gs`	`sr_gs`	2,469	0.11	0.54	0.0	0.930	yes
`era5`	`sr_era5`	2,965	0.44	0.35	0.0	1.000	yes

Figure 1 draws the record and the three sets on one clock, one record per panel, with the gaps left as gaps.

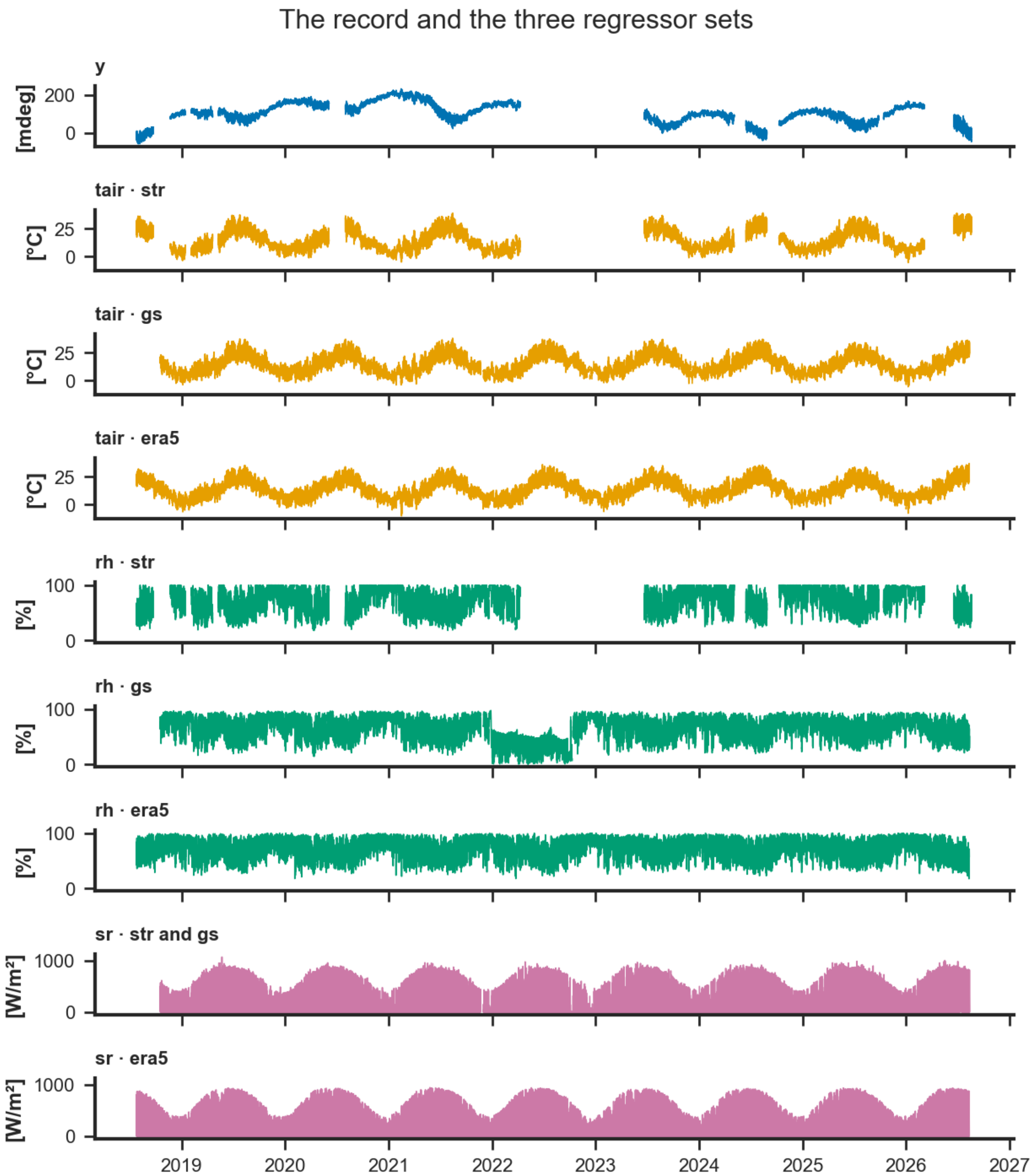

Figure 1: The record and the three regressor sets, 2018 to 2026, on the twenty-minute grid, one record per panel. Top: the compensated, once-anchored, spike-cleaned inclination, its gaps left as gaps. Below: air temperature, relative humidity and global horizontal radiation from each source, the panels of one role sharing their vertical scale. The station's radiation is drawn once and labelled for both the station set and the on-structure set that borrows it; the wall's own pyranometer does not enter the main line.

3 Method

Each subsection states one design decision, the evidence for it, and the alternative it rejects, in the order of the design document. Decisions whose results arrive in later phases, D6 onward, are stated here and reported in their own sections.

3.1 D1 · The whole record, 2018 to the archive end

Study 04's annual term was not identifiable on 3.2 cycles with a 103-day hole, and its trend disagreed with itself in sign across the thirds of its window: +38.6, +4.8 and +45.5 mdeg per year against −2.77 for the window as a whole. The whole record holds five usable annual cycles, 2019, 2020, 2021, 2024 and 2025, compensated and anchored once by Study 01. The instrument change of 21 February 2025 is not modelled: the recorded channel is continuous across it to +1.58 mdeg on thirty-day means, and the compensated gain is stable across both eras. The rejected alternative is a per-era offset, which would plant an arithmetic step the instruments did not record.

3.2 D2 · Three regressor sets through one model specification

One specification is fitted three times, once per regressor set. The on-structure set gives the site result, the station set the best external proxy, and the ERA5 set the transferable arm. The paper narrates two and tables all three. The on-structure set borrows the station's radiation, as Section 2 states. Wall temperature and on-structure radiation stay out of the main line, since they cover 17.6 % and 18.8 % of the window, and are tested on their own window in D4. The rejected alternative, a single set, would not separate what the site can achieve from what a deployment without on-structure instruments could.

3.3 D3 · Study 03's operator is applied outside the model; radiation is global horizontal in every set

Air temperature enters instantaneously. Radiation enters delayed by one hour, the diurnal-band optimum Study 03 measured for all three sources. Both pass through the library's one delay-and-filter operator, and no driver is given a time constant from the level band, where Study 03 showed such constants pin against the grid. Global horizontal irradiance is used in every set, so that the radiation gain is comparable across sets and matches the variable Study 03 screened. ERA5's skin temperature is not loaded: the radiative temperature of a grid cell several kilometres across, averaged over fields, roads and roofs, carries nothing about one stone face that the air temperature and radiation do not already carry. The rejected alternative, letting Model A learn the delay through lagged regressors, is the job of Model B in D9, where the learned response is confronted with the imposed one rather than replacing it.

3.4 D4 · Wall temperature and on-structure radiation, tested as a ladder on the current era

On the window from 21 February 2025 onward, the same specification is refitted rung by rung: the on-structure set as in the main line; on-structure radiation replacing the station's, on the certified days; the wall temperature added at zero delay with a four-hour time constant, the deployable form; and, as a diagnostic row only, the wall temperature at its measured lead of two hours, which is never used for the expectation. Each rung reports held-out error with a paired block-bootstrap increment, variance share, the learned gain beside Study 03's, conformal coverage and width, and residual autocorrelation. The verdict is one sentence per channel on whether a deployment should keep it. The rejected alternative, carrying both channels in the main line, would cost four fifths of the window.

3.5 D5 · The trend stays on in the monitoring fit

Study 04 removed the trend from its monitoring fit so that drift would remain in the residual. The consequence was a residual with lag-one autocorrelation of 0.995, a control limit swept out to 14.75 standard deviations, a 288-day alarm that said only that the wall had moved since calibration, and a detector blind to every slow mechanism. Here the expectation comes from a rolling refit every thirty days with the trend on; over thirty days the trend's extrapolation error is a fraction of a millidegree at the measured rates, and the residual it leaves is stationary enough to chart. Drift is detected by the slow chart of D10 on daily means, and by the sequence of refit trend slopes. Component shares, gains and the parameter figures come from a single attribution fit over the full training window.

3.6 D6 · Harmonic diagnostics before modelling

The Fourier orders of the seasonal terms and the weight curve of the conditional daily term are measured before any model is fitted, on two series: the target, and the residual of a plain least-squares regression of the target on the on-structure drivers, which is what the seasonal terms will actually have to explain. A Lomb–Scargle scan from twelve hours to nine hundred days certifies the periods; the daily cycle's amplitude and phase, fitted day by day, are then fitted against day of year with an annual Fourier series whose order is chosen on held-out years; and the singular values of the daily-by-annual surface say how many weighted daily shapes it takes to reproduce it. Every order and weight used afterwards traces to that table. The results are reported in Section 5.

3.7 D7 · Model A, an additive grey-box without autoregression

Model A is a linear trend with changepoints placed on covered time, an annual term and a daily term as Fourier series of the orders D6 measured, no weekly term, the three regressors of one set as future regressors, quantile regression at 0.05 and 0.95, and no autoregression, with the library's imputation switched off. The daily cycle is fitted as two shapes, a midsummer and a midwinter form, blended by two weights that vary only with the day of year and sum to one, so that the daily cycle's amplitude and phase modulate through the year without a boundary day at which the shape steps. The weight curve is the measured annual modulation of D6, normalised to the unit interval, with a cosine peaking in mid-July as the documented fallback. The conditional term is kept only if it improves the held-out error or carries more than one per cent of the variance, and four boolean seasons are the stated fallback if the smooth weights fail to fit. Training histories shorter than two full annual cycles are not allowed to speak.

3.8 D8 · Uncertainty the library's way, calibrated on a rolling window

Quantile regression inside the model gives a nominal ninety per cent interval; split conformal prediction then calibrates it. Study 04 calibrated once, on a stretch that ended eleven months before the record it covered, and reached 67.7 % coverage. Here each refit calibrates on the most recent six months of its own out-of-sample residuals, so the interval is never calibrated on a quieter era than the one it covers. Coverage, width, Winkler score and pinball loss are reported per set and per days since refit.

3.9 D9 · Model B, a learned impulse response on the on-structure set

The same specification without any autoregression on the target carries air temperature and the station's radiation as lagged regressors with thirty-six lags at twenty minutes, twelve hours of history per driver, at the resolution Study 03's scan could not reach. The learned weight per lag is the impulse response; from it an effective delay and a time constant are read and set beside Study 03's operator for each driver. Agreement validates the operator imposed in Model A; disagreement is reported, and Model A is not re-tuned to hide it. Phase 0 verified that the library accepts lagged regressors without autoregression on the target, so the hourly fallback the design provided for is not needed.

3.10 D10 · Three charts, one per damage mechanism and time scale

Reference statistics come from a fixed window, 2019 to 2021, before the 2022 outages and before the instrument change; the monitored record runs from January 2022. All three charts run on Model A's rolling residual for each set. The fast chart applies an exponentially weighted moving average and a CUSUM to the residual's prewhitened innovations at twenty minutes, for steps, spikes and instrument faults, at a budget of ninety watched days per false alarm. The daily chart applies the same pair to the amplitude and phase of a 24-hour harmonic fitted to each day's residual, for amplitude growth and phase change, at the same budget. The slow chart applies a CUSUM with a small reference value to the daily mean residual, beside the sequence of refit trend slopes, for drift, at a budget of one year. Every fast alarm is cross-checked against the air temperature, humidity and battery channels in the same slot, so that the episode table carries an attribution: environment, instrument, or unattributed.

3.11 D11 · Detectability by mechanism, with injections sized by the wall's daily response

Study 04's three injection shapes, amplitude growth, phase change and drift, are kept with two corrections. The phase injection is sized by the wall's fitted daily response on the injection date rather than by the leftover daily seasonal term, which on the measured numbers means about 5.6 mdeg for a two-hour shift where Study 04 injected 1.07. Injections are placed at several dates across the seasons, so that the detectability field is monotone rather than an artefact of one injection point. Each mechanism is scored on the chart built for it, with the other two reported as well, and an alarm counts only where the uncontaminated record is silent within the response window.

3.12 D12 · Outages as hypotheses, never as data

For each whole-day outage of seven days or more in the monitored period, the model refitted on the data up to the outage predicts the level at resumption from the station and ERA5 sets, which kept recording. The observed mean over the first seven days after resumption, with the restart transient excluded, is compared with the expected level and its conformal interval, giving a level shift with an interval and a verdict per outage and per proxy set. Nothing is written into the gap: an imputed value may serve as history or as hypothesis, never as evidence. Section 2 showed that the accepted target's gaps are longer than the archive's outages; which intervals are bridged is stated in Section 11.

3.13 D13 · The target is never filled; regressor dust is

Rows with a missing target are left out of every fit. Regressor gaps of at most two hours are filled by linear interpolation and flagged in a provenance column, because a regressor is a measured, smooth driver and not the thing being judged; longer regressor gaps leave the row out of the fit for that set only. Proxies are brought to the twenty-minute grid by linear interpolation between consecutive native observations, never across a longer gap, with the ERA5 radiation centred on the half-hour before interpolation. Each source's clock is checked before anything is joined, as Section 2 reports.

3.14 D14 · Every native plot type, redrawn in the project's graphical language

The library's native plots run in the notebook as diagnostics. The report shows the same content redrawn by the project's own figure functions under its binding graphical rules, each redraw reading its data through the library's public methods, the decomposed prediction, the trend and seasonal components, the lagged-regressor weights and the conformal output columns, through one extractor per plot type, and each extractor is tested against the numbers the native plot draws. Phase 0 found that this library version's matplotlib backend returns no figure object from its parameter plot; nothing in the design depends on it.

4 Trend

This section is written after Phase 2. Nothing here is a result yet.

5 Seasonality

5.1 The harmonic diagnostic

The Fourier orders of the seasonal terms and the weight curve of the conditional daily term were fixed before any model was fitted (D6). Two series were examined: the target itself, and the residual of an ordinary least-squares regression of the target on the three drivers of the on-structure set, which is what the seasonal terms will actually have to explain once the regressors have taken their share. That plain fit is a diagnostic device, not a model of the wall; on the levels it attributes the bulk of the daily and annual swing to air temperature; its coefficients are not reported here, and its residual carries whatever the drivers, entering instantaneously, do not.

Each series was scanned with a Lomb–Scargle periodogram, which takes the gaps as they are, on two bands ranked separately: from 0.4 to 3 days for the daily cycle and its harmonics, and from 30 to 900 days for the annual cycle and its harmonics (Table 4). The bands are kept apart because the sub-daily peaks carry a small share of the variance, about 1.1 % for the target's daily cycle and 0.1 % for the residual's, and a single ranking fills every rank with the long-period continuum before it reaches them. On the target the long band certifies the annual cycle at 364.2 days with a normalised power of 0.42, and the semi-annual cycle at 181.7 days with 0.035, inside its 11-day resolution of half a year; the remaining long-band entries at 311, 437 and 600 days are the leakage of the multi-year drift and of the record's gaps rather than cycles. On the residual the annual cycle survives at 352.7 days, within its 44-day resolution of a year, at a power of 0.061, and no semi-annual peak does: the drivers, entering instantaneously, take that component with them. The 900-day entry at the top of the residual's long band is the grid's edge, a third of the record, and is not a period. In the short band both series certify the daily cycle at 1.000 days and its 12-hour companion at 0.500 days; the entries at 0.9996, 1.0005 and 1.0013 days are the daily peak's sidelobes, spaced by the record's resolution and listed because they fall just outside the width within which the scan suppresses a peak's neighbours.

Table 4: Periods certified by the Lomb–Scargle scan on the target and on the regression residual, each band ranked against its own continuum. The resolution is the width around a period that this record cannot tell from the period itself. Read from `GM_04_harmonic_diagnostics.csv`.

Series	Band	Rank	Period [days]	Power
`target`	`short`	1	1.0000	0.0111
`target`	`short`	2	0.9996	0.0023
`target`	`short`	3	1.0005	0.0022
`target`	`short`	4	0.5000	0.0014
`target`	`short`	5	1.0013	0.0009
`target`	`long`	1	364.1820	0.4197
`target`	`long`	2	311.0393	0.0761
`target`	`long`	3	436.8641	0.0568
`target`	`long`	4	600.4277	0.0419
`target`	`long`	5	181.6826	0.0346
`residual`	`short`	1	0.9999	0.0012
`residual`	`short`	2	1.0005	0.0011
`residual`	`short`	3	0.5000	0.0007
`residual`	`short`	4	0.4999	0.0004
`residual`	`short`	5	0.9990	0.0004
`residual`	`long`	1	900.0000	0.0998
`residual`	`long`	2	583.3669	0.0666
`residual`	`long`	3	352.6925	0.0614
`residual`	`long`	4	453.1717	0.0433
`residual`	`long`	5	244.1000	0.0415

For every day with at least 60 of its 72 slots, a 24-hour harmonic and its 12-hour companion were fitted to the diurnal band of each series, giving the daily cycle's amplitude and the hour of its maximum over eight years (Figure 2). Each of those two daily series was then fitted against day of year with an annual Fourier series, the order chosen on held-out years and a higher order accepted only where it lowered the held-out error by at least one per cent (Table 5). On the target the fitted daily amplitude runs from a minimum of 2.8 mdeg in early December (day 342) to a maximum of 17.1 mdeg in late July (day 200), a six-fold swing; on the residual it runs from 2.2 mdeg in mid-April (day 102) to 3.3 mdeg in late July (day 206), a swing of one and a half. The two share the summer peak and not the trough: once air temperature has been removed, the residual's daily cycle is smallest in spring rather than in winter. Both fits take order two, and the residual's fit, the one the conditional daily term consumes, has coefficients $(2.620, -0.292, -0.135, 0.265, 0.268)$ for the constant, the first cosine and sine and the second cosine and sine over a period of 365.25 days; normalised to the unit interval it peaks on day 206. That the residual's daily cycle modulates weakly once air temperature has been removed is itself a finding: most of the annual modulation of the daily swing rides on the driver, and the conditional term of D7 has less to carry than the target's picture suggests. Whether the two weighted shapes still earn their place is decided by the held-out comparison reported in the second half of this section.

Table 5: Annual modulation of the daily cycle's amplitude and hour of maximum, per series: the leave-one-year-out error of each candidate Fourier order and the order chosen. Read from `GM_04_harmonic_diagnostics.csv`.

Series	Statistic	Order	Held-out MSE	Chosen
`target`	`amplitude`	1	14.577	no
`target`	`amplitude`	2	13.248	yes
`target`	`phase`	1	3.581	yes
`target`	`phase`	2	3.583	no
`residual`	`amplitude`	1	2.880	no
`residual`	`amplitude`	2	2.805	yes
`residual`	`phase`	1	9.820	no
`residual`	`phase`	2	9.486	yes

The hour of the daily maximum is a circular quantity, and it was recentred on each series' circular mean before fitting, so that the two labels of midnight did not pull the fit apart. On the target the maximum sits near midnight throughout the year and its annual fit takes order one, with a held-out error that order two does not improve: the inclination's daily cycle keeps its timing, the wall tipping farthest towards the mountain in the afternoon and recovering overnight. On the residual the maximum sits near 17 h in winter and moves to about 23 h in July, an excursion of about six hours that order two fits better than order one by three per cent held out. What the drivers leave behind therefore changes its shape as well as its size through the year, which is the case for two weighted daily shapes rather than one scaled shape.

The diurnal band of the residual, binned by week of the year and slot of the day, is a surface whose singular values say how many weighted daily shapes it takes to reproduce it (Table 6). The first component carries 69 % of its variance and the second 24 %, 92.7 % together; a third adds 5 %. Two blended shapes therefore reproduce the surface to within a few per cent, which licenses the two conditions of D7 and does not call for a third.

Table 6: Singular-value decomposition of the residual's daily-by-annual surface: share of variance per component and cumulative share. Read from `GM_04_harmonic_diagnostics.csv`.

Component	Share	Cumulative
1	69.1%	69.1%
2	23.6%	92.7%
3	5.0%	97.8%
4	0.7%	98.4%
5	0.5%	99.0%
6	0.2%	99.1%

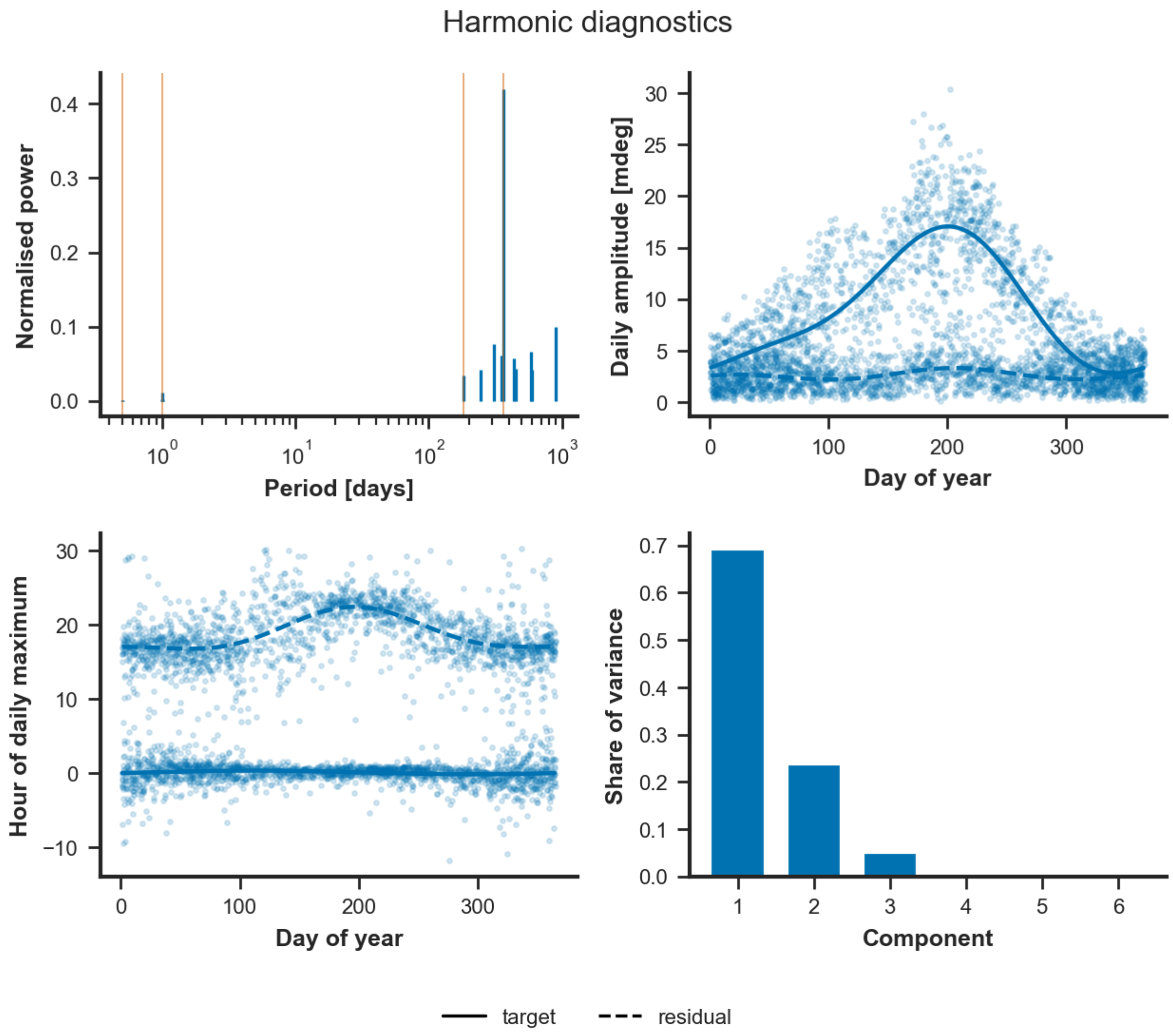

Figure 2: The harmonic diagnostic. Top left: the periods the scan certifies on the target (solid) and on the regression residual (dashed), on a logarithmic period axis, with the half-day, day, half-year and year marked. Top right: the daily amplitude by day of year with its annual fit. Bottom left: the hour of the daily maximum, recentred on its circular mean, with its annual fit. Bottom right: the share of variance of the leading components of the daily-by-annual surface.

Three parameters follow from the diagnostic and are declared in the notebook's parameter cell with a pointer to `GM_04`. The yearly term takes Fourier order one, because the residual certifies the annual cycle and no semi-annual one. The daily term takes order two, because both series certify the 12-hour companion of the daily cycle, the signature of a daily response that heats faster than it cools. The weight curve of the conditional daily term is the residual's order-two annual fit of the daily amplitude, normalised to the unit interval and peaking on day 206, and the two weighted shapes go forward to Phase 2's held-out test, since the first two components of the surface carry 92.7 % of its variance.

This section is written after Phase 2 — the fitted yearly and daily terms. Nothing here is a result yet.

6 Regressors and gains

This section is written after Phase 2. Nothing here is a result yet.

7 Impulse response

This section is written after Phase 4. Nothing here is a result yet.

8 Uncertainty and validation

This section is written after Phase 3. Nothing here is a result yet.

9 What the wall temperature and the pyranometer buy

This section is written after Phase 2b. Nothing here is a result yet.

10 The monitor

This section is written after Phase 5. Nothing here is a result yet.

11 Outages as hypotheses

This section is written after Phase 6. Nothing here is a result yet.

12 Verdict

This section is written after Phase 7. Nothing here is a result yet.

13 Limitations

This section is written after Phase 7. Nothing here is a result yet.

14 Run metadata

This section is written after Phase 7. Nothing here is a result yet.